Forget-Set Composition, BatchNorm Handling, and the Stability of NegGrad+: Exploratory Diagnostics on Fashion-MNIST

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Manuscript draft for advisor feedback · 13 October 2026 · not for circulation

All analyses after the original v3 study are post hoc and exploratory. No privacy or erasure claim is made.

Abstract

We report a retrospective account of exploratory experiments on approximate unlearning in a small convolutional network with three BatchNorm layers trained on Fashion-MNIST. The original study (v3) compared two kinds of forget sets of identical size (600 examples): a stratified random set (60 per class) and class-concentrated subsets (600 examples from one class, whose remaining examples stay in the retained data). It used five evaluation seeds, a retrained-from-scratch reference, fine-tuning on retained data, and NegGrad+ implemented with separate retain and forget forward passes in training mode.

Fine-tuning showed no detectable difference between concentrated and random forget sets in the gap to retraining on target-class test accuracy (mean paired difference -0.74 points, 95% CI $[-2.33, 0.86]$, 5 seeds). NegGrad+ satisfied the study's prespecified directional criterion. However, it did so through runs whose test accuracy fell to chance level: NegGrad+ ended at $\leq 20\%$ test accuracy in 26 of 55 evaluation runs, all with concentrated forget sets.

Post hoc diagnostics showed the following:

- BatchNorm running buffers strongly changed the measured accuracy of these runs: medians of 10.0% versus 68.9–74.1% for identical weights evaluated with different buffers. Buffer updates from forget forward passes alone did not produce collapse in an $\alpha = 1$ control (0/55).
- A concatenated retain+forget forward pass and BatchNorm in evaluation mode during unlearning both avoided divergence in all 30 concentrated tasks examined, where the original implementation diverged in 29.
- Further interventions on the forget forward pass gave mixed results. Normalizing it with current running buffers avoided divergence; detaching batch statistics in the backward pass reduced but did not prevent threshold crossing; using the original model's frozen buffers diverged even on a control class.

The stable variants stayed close to the original model and to fine-tuning on the forgotten examples. These results concern optimization and behavior only. They do not establish successful forgetting, erasure, or privacy.

1 Introduction and research question

Approximate unlearning methods are usually compared against a model retrained without the data to be forgotten. Several recent works report that the composition of the forget set changes how hard unlearning is [4, 6]. They also report that BatchNorm statistics can distort the evaluation of unlearned models [1].

Our starting question was narrow. At a fixed forget-set size, does concentrating the forget set in one class, while that class remains in the retained data, change the apparent effectiveness of

approximate unlearning, and its behavioral proximity to retraining? We call this regime *instance deletion*, as opposed to removing a whole class.

The original study produced an unexpected result: NegGrad+ often collapsed to chance-level accuracy when the forget set was concentrated. The study therefore turned into a sequence of post hoc diagnostics. These diagnostics ask which changes to the training-mode forward pass on the forget batch, and to BatchNorm handling more generally, are associated with that instability.

We report this sequence in the order in which it was run, and label which analyses were fixed in advance. The evidence is exploratory: one small architecture, one dataset, frozen hyperparameters and 5 seeds. We make no claim that a general mechanism has been identified.

2 Related work (preliminary review)

This review is preliminary and not exhaustive. Full texts and source code were read with the help of a summarising web-fetch tool, so quoted wording must be re-checked against the PDFs before submission. We make no novelty claim. Bibliographic details are limited to what was verified, and no acceptance status is asserted.

BatchNorm artifacts in unlearning evaluation. Kalani et al. [1] show that forward passes in training mode over forget data shift BatchNorm running statistics toward the forget class. This can make forget accuracy look low; recalibrating the statistics on retained data with weights fixed reverses much of the apparent forgetting.

- *Their setting:* CIFAR-10/100 with ResNet-18, plus larger-scale checks; mostly class-level forgetting, with one instance-wise condition.
- *Controls:* GroupNorm and LayerNorm, for which the recalibration gap is zero.
- *Composition:* mixing retain data into the unlearning batches, or forgetting random instances, reduces the artifact.
- *What they do not cover:* NegGrad+ is not evaluated, and we found no report of divergence or non-finite losses.

We use their recalibration idea. Our object differs: optimization stability of NegGrad+ under partial-class deletion compared with a size-matched stratified set.

PURGE [2] reports, as a debugging observation during class-level unlearning with ResNet-18, that class-homogeneous forget batches overwrote BatchNorm statistics: early CIFAR-10 runs collapsed to 21.6% test accuracy, and switching BatchNorm to evaluation mode resolved it. Its setting is full-class forgetting, without a composition-controlled comparison.

NegGrad+ implementations. We inspected the source code of three repositories. The table describes the code we read, which is not necessarily the configuration used in each paper's experiments.

Our original implementation shares RUM's BatchNorm handling, and our variant C shares SCRUB's evaluation mode, but neither matches the other choices of those codebases.

Forget-set difficulty. Zhao et al. [4] relate unlearning difficulty to the entanglement of forget and retain sets and to memorization; their forget sets span all classes. Fan et al. [6] search for worst-case forget sets by bi-level optimization. Neither studies class-concentrated partial deletion against a size-matched stratified set, nor BatchNorm handling.

Instability of gradient ascent. Zhang et al. [7] describe a catastrophic collapse of gradient-ascent unlearning in language models. Mavrothalassitis et al. [8] report that gradient ascent and gradient descent-ascent either do nothing or break the model in image classifiers, depending

Repository and function	Forward passes	Mode during unlearning	Other choices
SCRUB [3], `run_neggrad_epoch` in `small_scale_unlearning.ipynb`	Retain and forget passed separately	`model.eval()`	SGD without momentum; loss includes an L2 penalty toward the initial weights. The paper text remarks that small β makes the optimization explode quickly
RUM [4], `unlearn/neggrad.py`	Separate	`model.train()`	Batch size 256; momentum 0.9
SalUn [5]	Forget data only (plain gradient ascent)	`model.train()`	No NegGrad+ variant found

on how forget and retain sets depend statistically. Their forget sets are defined by influence directions, not by class.

BatchNorm with non-i.i.d. batches.

- Batch Renormalization [9] shows that class-structured minibatches degrade BatchNorm training.
- Wu and Johnson [10] discuss inconsistency between training and test statistics, and information leakage through batch statistics.
- Federated learning (FedBN [11]) and continual learning (Continual Normalization [12]) document running statistics biased toward the current data.

None of these sources tests unlearning.

Evaluation. Hayes et al. [13] argue that common membership-inference evaluations of inexact unlearning overstate privacy protection. We ran no membership-inference attack.

Novelty boundary. In our non-exhaustive search, we found no verified work comparing class-concentrated partial deletion with a size-matched stratified forget set in terms of NegGrad+ optimization stability, or manipulating BatchNorm handling of the forget forward pass in that setting. This is a statement about our search, not a novelty claim.

3 Experimental setup

3.1 Data, forget sets and retained sets

- **Splits.** From the 60,000 Fashion-MNIST training images, a stratified validation set of 5,000 (500 per class) was held out with a fixed seed. This leaves $D = 55{,}000$ images (5,500 per class). The 10,000 test images were used only for evaluation.
- **Conditions.** Each evaluation seed defines 11 forget sets of size $n = 600$ drawn from D: one stratified random set with 60 per class, and one concentrated set per class with 600 examples of that class (10.9% of it). The remaining 4,900 examples of that class stay in the retained set D_r. Conditions are named `alea` (stratified) and `c0`–`c9`.
- **Seeds.** Seeds 1–5 were used for evaluation, seed 0 for tuning and seed 1000 for a technical pilot that is excluded from all analyses.

3.2 Model and reference models

The network has the layers Conv(32)–Conv(64)–MaxPool–Conv(128)–MaxPool–FC(256)–FC(10), with BatchNorm after each convolution (three BatchNorm layers) and about 1.70 M parameters. The original model O and the retrained models R (one per seed and condition, trained on D_r) used the same recipe: SGD with momentum 0.9, learning rate 0.05 with cosine decay, weight decay $5 \cdot 10^{-4}$, batch size 128, 15 epochs and no augmentation. Test accuracy of O was 93.03–93.37% for seeds 1–5 (93.07% for seed 0).

R serves as a *behavioral reference*. It is trained without the removed indices, but this does not show that information present elsewhere in the data, or learned generally, is absent.

3.3 Unlearning methods

- **Fine-tuning (FT)** on D_r, with SGD (momentum 0.9, weight decay $5 \cdot 10^{-4}$) at a constant learning rate in $\{0.01, 0.05\}$, for $E \in \{1, 3, 5\}$ epochs.
- **NegGrad+** minimizes $\alpha \cdot \mathrm{CE}_r - (1-\alpha) \cdot \mathrm{CE}_f$, where CE_r is the cross-entropy on retained data and CE_f on forgotten data.
  - Optimizer: learning rate 0.01, the same momentum and weight decay, gradient-norm clipping at 5, $\alpha \in \{0.9, 0.95, 0.99\}$, $E \in \{1, 3\}$.
  - One epoch is one pass over D_r in batches of 128 (425 steps); each step pairs a retain batch with a forget batch of 64 drawn from a continuous stream of permutations of the forget set.
  - The *original implementation*, denoted **A** (and **V0** in diagnostic 1), passes the retain and forget batches through the network separately, in training mode.

3.4 Tuning rule (v3, without the retrained model as oracle)

For each condition and method, a configuration was chosen on the tuning seed (seed 0) using only the validation set and O:

- preservation $P = \mathrm{Acc}_{\mathrm{val}}(O) - \mathrm{Acc}_{\mathrm{val}}(U) \leq \tau = 1$ point;
- forgetting gap $G = |\mathrm{Acc}_F(U) - \mathrm{Acc}_{V_k}(U)|$, where V_k is the validation subset matching the composition of the forget set;
- a binomial tolerance δ_k defines a set of configurations equivalent in G, and the least aggressive one in that set is chosen;
- if no configuration is admissible, the one minimizing P is used, and this fallback is flagged.

The selected configurations were then frozen for seeds 1–5:

- NegGrad+: $\alpha = 0.99$, $E = 1$ for every concentrated class, and $\alpha = 0.95$, $E = 1$ for the stratified set;
- FT: lr 0.05 or 0.01, with $E = 1$ or 3, depending on the condition.

The fallback was triggered for NegGrad+ in classes c1, c2, c4, c5, c6 and c7. In five of these, the selected configuration had $P = 83.42$ points, that is a model at about chance accuracy on validation; in c6, $P = 11.0$.

This exposes a weakness of the rule: a constant predictor gets $G = 0$. It also introduces a confound, since the stratified condition used a larger forget weight ($1 - \alpha = 0.05$) than the concentrated ones (0.01).

3.5 v3 hypotheses and statistical criteria

Two estimands were defined for each seed and method, averaged over the 10 classes c:

- **H1**: the mean over classes of [drop in accuracy on the 600 forgotten examples of c in the concentrated condition − drop on the 60 examples of c in the stratified condition], each drop being measured relative to O.
- **H2**: the mean over classes of [$|\mathrm{Acc}_{T_c}(U) - \mathrm{Acc}_{T_c}(R)|$ in the concentrated condition − the same quantity in the stratified condition], where T_c is the test set of class c.

The decision rule, across 5 seeds with a two-sided Student 95% CI (4 degrees of freedom):

- "supported" if the mean is > 0, the CI excludes 0 and every seed is > 0;
- "contradicted" if the CI is entirely < 0;
- otherwise "inconclusive".

The executed v3 notebook shows that a preregistration file was written to the project drive, in the same session, before the main-experiment cell ran.

[OPEN CHECK: the dated preregistration file itself was not inspected for this draft; confirm its timestamp and wording.]

3.6 Divergence, collapse and utility (used from diagnostic 1 onward)

- **Collapse**: test accuracy $\leq 20\%$ with the model's own buffers.
- **Divergence**: maximum per-step CE_f during unlearning > 10, or a non-finite loss. Used from diagnostic 2 onward.
- **Utility retained**: test accuracy at least that of O minus 5 points, both with the model's own buffers and after recalibration. Used from diagnostic 2 onward.

Divergence and collapse are different outcomes and are reported separately. All thresholds are operational conventions.

3.7 Reproducibility

All runs used an NVIDIA A100-SXM4-40GB with Python 3.13.15, PyTorch 2.11.0+cu130 and CUDA 13.0. cuDNN ran in deterministic mode with `torch.use_deterministic_algorithms(True, warn_only=True)` and a fixed cuBLAS workspace. The v3 pilot found retraining bitwise reproducible. Each diagnostic re-ran the original NegGrad+ implementation and reproduced the stored v3 predictions bitwise on its checked tasks.

4 Results

4.1 Original study (v3)

PRESPECIFIED ANALYSES

The H2 values for each seed were recomputed from the per-class raw counts (`h_par_classe.csv`) and match the summary file.

Fine-tuning. We detected no difference between conditions. Fine-tuned test accuracy ranged from 90.0% to 93.2% across the 55 evaluation runs.

Table 1: v3 hypotheses, 5 seeds. Values are in percentage points; source `T2_h1_h2_resume.csv`.

Hyp.	Method	Mean	95% CI	Per-seed values	Prespecified decision
H1	Fine-tuning	−0.34	[−3.76, 3.08]	0.65; −2.57; −1.98; 4.07; −1.87	inconclusive
H2	Fine-tuning	−0.74	[−2.33, 0.86]	0.46; −2.02; −1.37; 0.82; −1.57	inconclusive
H1	NegGrad+	53.19	[45.72, 60.65]	55.85; 55.97; 52.70; 43.05; 58.37	"supported" (criterion only)
H2	NegGrad+	54.15	[47.84, 60.45]	57.48; 54.89; 53.17; 46.05; 59.14	"supported" (criterion only)

NegGrad+. The prespecified criterion was satisfied, but the result reflects optimization failure rather than useful unlearning. Table 2 shows NegGrad+ test accuracy for each run. Twenty-six of 55 runs ended at $\leq 20\%$; none of the stratified runs and none in c0, c3, c8 or c9 did.

Table 2: Test accuracy (%) of NegGrad+ in the v3 evaluation runs (seeds 1–5). Source `runs.csv`. Values ≤ 20 are highlighted.

Condition	s1	s2	s3	s4	s5
alea (stratified)	92.4	92.6	91.9	92.2	92.3
c0	91.9	92.8	92.8	92.8	89.1
c1	**10.0**	**10.0**	**10.0**	**10.0**	**10.0**
c2	**10.0**	**10.0**	**10.0**	**10.0**	**10.0**
c3	89.9	92.3	92.4	92.4	90.8
c4	**10.0**	**10.0**	**10.0**	**10.0**	**10.0**
c5	**10.0**	**10.0**	63.7	90.9	**10.0**
c6	**10.1**	**10.0**	**10.0**	**10.0**	**10.0**
c7	20.9	**16.9**	**11.9**	**12.4**	31.2
c8	92.4	92.8	92.9	92.8	92.7
c9	92.6	92.7	92.7	93.1	92.3

4.2 Diagnostic 1: running buffers versus weights

POST HOC

Design. For all 55 evaluation tasks, NegGrad+ was re-run with the frozen v3 configurations, reproducing the v3 predictions bitwise. Variants were run in lockstep on identical batches:

- **V0**: the original implementation;
- **V1a**: BatchNorm momentum 0 on both forward passes, so running buffers stay those of O;
- **V1c**: momentum 0 only on the forget forward pass;
- **V2**: V0's weights with buffers recalibrated on D_r. Buffers were reset, then accumulated as a per-batch cumulative average over D_r, in training mode, without gradients and with weights fixed.

Because training-mode BatchNorm normalizes with batch statistics, V0, V1a and V1c have identical parameters at every step; per-step assertions enforced this. A control with $\alpha = 1$ kept the forget forward passes but gave them zero loss weight. Its parameters were asserted equal at every step to those of a fine-tuning run on the same retain batches.

Results.

- V0 collapsed (test $\leq 20\%$) in **26 of 55** tasks: c1, c2, c4 and c6 on 5/5 seeds, c5 on 3/5 and c7 on 3/5. No stratified task collapsed.
- Over these 26 collapsed tasks, median test accuracy was:
  - V0: 10.00%;
  - V1a (original buffers): 10.00%;
  - V1c (buffers updated by retain passes only): 68.91%;
  - V2 (recalibrated): 74.10%.
- The $\alpha = 1$ control never collapsed (0/55; minimum test accuracy 91.42%).
- Recalibrating O and R changed their test accuracy by at most 0.04 and 0.09 points.

Interpretation. Running buffers strongly affected the measured accuracy of these runs. Buffer contamination from forget forward passes alone did not produce collapse in this setup. Even the best buffers left accuracy far below that of O (about 93%), so the learned weights were also damaged. According to the per-task summaries (`lecture_par_tache.csv`), the c1 runs reached maximum forget losses of $1.9\text{–}3.0 \cdot 10^5$, and 68–74% of their steps were clipped.

[The full per-step traces are stored on the project drive and were not inspected for this draft.]

4.3 Diagnostic 2 and its extension: forward composition and BatchNorm mode

POST HOC

Design. The reading criteria were written before this diagnostic was computed, but the conditions were chosen from observed results. It compared, on identical batches:

- **A**: the original implementation;
- **B**: one forward pass on the concatenated retain+forget batch, with losses computed separately on the two slices;
- **C**: all BatchNorm layers in evaluation mode during unlearning, with O's buffers and trainable γ, β;
- **D**: A with $\alpha = 1$;
- **FT-eq**: fine-tuning on the same retain batches.

B changes more than the statistics used for normalization: it also creates interactions between examples within the batch, and updates the buffers once per step instead of twice. C changes normalization during optimization, and therefore the gradients and weights. Neither comparison isolates a single mechanism.

Table 3: Diagnostic 2, 5 seeds per condition. Source `comptes_par_condition.csv` and `variantes_par_tache.csv`.

Condition	Variant	Divergence	Collapse	Median test acc. (%)	Median max CE_f
c1	A	**5/5**	**5/5**	10.00	$2.86 \cdot 10^5$
c1	B	0/5	0/5	92.84	0.051
c1	C	0/5	0/5	92.46	0.093
c1	D ($\alpha = 1$)	0/5	0/5	92.27	0.320
c8 / alea	A, B, C, D	0/5 each	0/5 each	92.06–92.85	≤ 1.64

Extension. B and C were run on c2, c4, c5, c6 and c7, 5 seeds each, with references for A reused from earlier diagnostics. The step-0 losses of A were checked bitwise against those references for all 25 tasks, and A was replayed in full on one task.

- Including c1 from diagnostic 2, A diverged in **29 of 30** concentrated tasks; B and C diverged in **0 of 30**.
- B and C kept utility in all 50 extension runs (test accuracy 91.80–93.34%).
- Per class, A diverged on c2, c4, c6 and c7 in 5/5 seeds and on c5 in 4/5. Under the prespecified class rule, c5 is *indeterminate* because A did not diverge on all seeds; it is reported seed by seed.
- Divergence is not the same as collapse. Three A runs crossed the divergence threshold without collapsing:
  - c5, seed 3: max CE_f 32.1, test 63.7%;
  - c7, seed 1: max CE_f $5.1 \cdot 10^5$, test 20.86%;
  - c7, seed 5: max CE_f $5.4 \cdot 10^5$, test 31.18%.
- One A run (c5, seed 4) neither diverged nor collapsed: max CE_f 1.22, test 90.9%.

Table 4: Accuracy (%) on the forgotten examples, medians over 5 seeds (extension). The original model O scores 99.3–100 on these examples. Source `par_classe_graine_variante.csv` and v3 `metriques_long.csv`.

Class	R (retrained)	B	C
c2	91.67	97.50	95.50
c4	90.00	96.33	96.17
c5	98.67	99.83	100.00
c6	79.50	94.50	93.00
c7	97.17	99.17	98.50

On the forgotten examples, B and C stayed closer to O than to R. Their behavior resembles that of fine-tuning (FT-eq $\approx$ 100% on c1), so they show little effect of the ascent term in these settings.

4.4 Diagnostic 3: interventions on the forget forward pass (c1, c8)

POST HOC

Design. The retain forward pass is always identical to A's; only the forget forward pass changes.

- **E2**: training-mode batch statistics, without buffer updates. This is identical to V1c and reused from diagnostic 1.
- **E1**: normalization with the current running buffers, which reflect retain batches only. This changes both the statistics used and the backward pass.
- **E1-O**: normalization with O's frozen buffers.
- **E1-detached**: the same forward pass as A, with batch statistics computed using biased variance and the same ε, but the mean and variance are detached in the backward pass. This changes only the backward pass and is not standard BatchNorm.

342 named consistency checks passed. They include bitwise reproduction of A's references, per-step equality of E2 with A in the replayed pilot, and unit tests of the detached forward and backward passes.

Table 5: Diagnostic 3, by seed. Max CE_f during unlearning / test accuracy with own buffers (%). Values for A and E2 are references; E2 has A's weights and differs only by its buffers. Source `resultats_par_tache_variante.csv`.

Class	Variant	s1	s2	s3	s4	s5
c1	A	**2.9e5 / 10.0**	**2.9e5 / 10.0**	**2.2e5 / 10.0**	**3.0e5 / 10.0**	**1.9e5 / 10.0**
c1	E2	= A / 71.0	= A / 72.8	= A / 74.8	= A / 69.1	= A / 79.2
c1	E1	0.043 / 92.4	0.024 / 93.3	0.035 / 93.0	0.046 / 92.8	0.078 / 92.8
c1	E1-detached	11.3 / 88.0	33.6 / 41.2	23.2 / 33.5	12.4 / 32.2	13.8 / 49.8
c1	E1-O	**3.7e10 / 10.1**	**6.8e13 / 10.0**	**5.0e14 / 10.0**	**1.9e16 / 10.0**	**1.5e16 / 10.0**
c8	A, E2, E1, E1-detached	No divergence (max $CE_f \leq 0.33$); test accuracy 92.4–93.0				
c8	E1-O	**6.3e11 / 9.6**	**2.0e14 / 9.0**	16.9 / 74.2	**3.1e16 / 10.2**	**4.5e11 / 10.1**

Observations.

- **E1-detached** crossed the divergence threshold of 10 on c1 in 5/5 seeds, but with maxima of 11.3–33.6 rather than about 10^5. Three of these maxima are only slightly above the threshold. Divergence started at steps 63–99 of 425. None of these runs collapsed below 20%, but none retained utility (test 32.2–88.0%).
- **E1-O** diverged on all c1 seeds and also on all c8 seeds, collapsing on 4/5 of the latter.

Prespecified readings:

- E2 is ruled out as a cause of divergence.
- The effect of batch statistics, assessed with E1 and E1-O together, is *indeterminate*: 0/5 versus 5/5.
- The gradient through the statistics, assessed with E1-detached, is *weakened* as a necessary cause of threshold crossing.
- The c8 control flags E1-O as a variant-specific instability.

Post hoc pairwise comparisons, not preregistered:

- *A versus E1-detached* (same forward pass, different backward pass): removing the gradient path through batch statistics coincided with a reduction of several orders of magnitude in max CE_f, but not with avoiding the threshold.
- *E1 versus E1-detached* (both without a gradient through the statistics; they differ in the statistics used): using the forget batch's own class-homogeneous statistics in the forward pass coincided with threshold crossing.
- *E1-O*: frozen statistics from O destabilized even c8. This complicates any simple "fixed versus adaptive statistics" account.

5 Discussion

We keep five notions separate:

1. **Optimization divergence**, meaning a runaway forget loss. In this implementation and these settings, separate training-mode forward passes on class-concentrated forget batches were associated with divergence in 29 of 30 concentrated tasks, and never with the stratified set (0/5 tasks; the same five runs were reproduced bitwise in v3 and diagnostics 1–2). The stratified condition also used a different frozen α.

2. **Classification collapse**: test accuracy $\leq 20\%$. It usually, but not always, accompanied divergence: three diverged runs did not collapse.
3. **Retained utility**. The concatenated (B) and evaluation-mode (C) variants, and E1, kept test accuracy within about 1.5 points of O (91.8–93.3% against 93.0–93.4%). E1-detached kept neither its utility nor its stability.
4. **Behavioral proximity to retraining.** In partial-class deletion, R remains close to O on most metrics. The stable variants were at least as close to O as to R on the forgotten examples, and behaved like fine-tuning. Where O and R differ most (c6: 99.3% versus 79.5% on the forgotten examples), B and C covered only a small part of that gap.
5. **Formal erasure and privacy** were not assessed. No membership-inference attack was run, and recalibrated evaluations measure sensitivity to buffers, not erasure.

The v3 result for NegGrad+ illustrates a reporting hazard: a prespecified directional criterion can be satisfied by runs that destroy the model. We therefore report NegGrad+ in v3 as a *failure (instability) of the method under this grid and implementation*, not as evidence of effective forgetting.

Three findings bear on mechanism:

- Buffer updates did not change the training trajectory, and the $\alpha = 1$ control did not collapse. Buffer contamination therefore does not account for the optimization collapse on its own, although buffers strongly affected measured accuracy.
- B, C and E1 all changed how the forget batch is normalized, and all avoided divergence. E1-detached and E1-O did not.
- These are associations under specific interventions; they do not identify a general mechanism.

6 Limitations

- **Scope:** one small CNN with three BatchNorm layers, one dataset (Fashion-MNIST), forget-set size $n = 600$ (10.9% of a class), and 5 evaluation seeds.
- **Tuning:** configurations were frozen from a single tuning seed, and the tuning fallback selected collapsed configurations for six classes. The stratified and concentrated conditions also used different frozen α values.
- **Post hoc selection:** diagnostics 1–3 and the extension are post hoc. Their classes (c1, c8, then c2–c7) were selected after observing v3, although the per-diagnostic reading criteria were written before each computation.
- **Reference model:** one retrained model per condition, with no estimate of retraining variability.
- **Recalibration:** diagnostics 1–3 used per-batch cumulative averaging. This differs from the population-weighted, sequential procedure specified in the later replication protocol.
- **Thresholds and verification:** the thresholds (divergence > 10, collapse $\leq 20\%$, utility at O $-$ 5 points) are conventions. For the extension and diagnostic 3, the batch plans of the reused references were verified at step 0 for every task and in full only on replayed pilot tasks.
- **Literature:** the review is preliminary.

7 Reproducibility notes

Notebooks:

- `unlearning_structure_oubli_fmnist.ipynb` (v3);
- `diagnostic_bn_neggrad.ipynb` (diagnostic 1);
- `diagnostic_bn2_normalisation.ipynb` (diagnostic 2);
- `diagnostic_bn2_extension.ipynb` (extension);
- `diagnostic_bn3_statistiques.ipynb` (diagnostic 3).

Results are stored in the project drive folder `unlearning_fmnist_v3/` and its subfolders `diagnostic_bn/`, `diagnostic_bn2/`, `diagnostic_bn2_ext/` and `diagnostic_bn3/`.

Environment: A100, PyTorch 2.11.0+cu130 and deterministic settings in all sessions. The main v3 experiment took 34.9 minutes. Each diagnostic stored named consistency checks and refused to analyze if any check failed.

8 Conclusions

In a small BatchNorm CNN on Fashion-MNIST, NegGrad+ implemented with separate training-mode forward passes became unstable for most class-concentrated forget sets of 600 examples, and was stable for a size-matched stratified set. A prespecified criterion was met only through these failures.

Concatenating the retain and forget batches, or running BatchNorm in evaluation mode during unlearning, avoided divergence in every concentrated task we tested. These stable variants changed the model little beyond fine-tuning.

The evidence is exploratory and concerns optimization and behavior only. It does not establish successful forgetting, erasure or privacy, and it does not identify a general mechanism.

Future work (proposed, not run)

A dated replication protocol (v1.3, 13 October 2026) proposes:

- a principal Fashion-MNIST study comparing BatchNorm implementations A and C on all ten classes, with three forget-loss intensities and five seeds;
- optional, separately analyzed replications with GroupNorm (a different system) and with CIFAR-10 and ResNet-18;
- a failure-inclusive ordinal estimand, and population-weighted sequential recalibration.

A technical pilot notebook has been prepared. **None of this has been executed**, and no planned quantity in that protocol is a result.

9 Proposed tables and figures

- **Table 1**: v3 hypotheses with per-seed values (above; available).
- **Table 2**: v3 NegGrad+ test accuracy by class and seed (above; available).
- **Table 3**: tuning on seed 0, with P, G and the selected configuration per condition, flagging the six fallback selections (available: `T3_configs_retenues.csv`, `tuning.csv`).
- **Table 4**: diagnostic 1 medians over the 26 collapsed tasks for V0, V1a, V1c and V2, plus the $\alpha = 1$ control (available).
- **Table 5**: diagnostic 2 and extension, divergence and collapse for A, B and C by class and seed (available).

- **Table 6**: diagnostic 3 by seed (above; available).
- **Figure 1**: CE_f trajectories for A, B, C, E1, E1-detached and E1-O on c1 versus c8.

  [Traces stored on drive (`diagnostic_bn2/traces`, `diagnostic_bn3/traces`); not inspected for this draft.]
- **Figure 2**: test accuracy with the model's own versus recalibrated buffers, per variant (available from the result CSVs).
- **Figure 3**: accuracy on the forgotten examples for O, R, FT, B and C by class (available).
- **Figure 4**: decomposition of retain and forget gradient norms over training.

  [Data in `*_periodiques.csv` on drive; not inspected.]

10 Open checks before submission

1. Inspect the dated v3 `preregistration.md` and `preregistration_params.json` on the drive, to confirm their timestamp and wording relative to the main-experiment run. This draft relies on notebook execution order only.
2. The v3 protocol versions v1–v3 were drafted in a working session; there is no dated local copy other than the drive file mentioned in item 1.
3. Inspect the per-step traces and periodic gradient logs (diagnostics 1–3) before reporting divergence onset for A, gradient decomposition or per-layer drift. The values quoted here come only from summary CSVs.
4. The per-task JSON files of diagnostic 1, which carry the per-step parameter-equality flags, were not inspected directly. Their status is inferred from completed runs and from a passed cross-check in diagnostic 3.
5. Verify every literature quotation and bibliographic field against the PDFs, including the full author lists of [3] and [5]. Do not state venues until verified.
6. Decide whether to report the confound that the stratified condition used $\alpha = 0.95$ while all concentrated conditions used $\alpha = 0.99$, and whether a matched-α analysis is needed.
7. Retraining variability (R versus R′) was not measured. Proximity statements depend on a single R per condition.

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